

Project Design Phase
Proposed Solution

Date	05-07-2026
Team ID	SWTID-2026-8108
Project Name	Heart Disease Analysis
Maximum Marks	

Proposed Solution :

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Crucial medical and lifestyle risk data is trapped in 4,416 rows of relational text tables, making it too slow and complex for clinicians and analysts to find critical patient risk trends without manual database coding.
2.	Idea / Solution description	An interactive, tiled Tableau dashboard connected directly to a local MySQL database engine that provides 1-click cross-filtering across age, BMI, smoking status, and regional demographics on a single screen.
3.	Novelty / Uniqueness	Multi-perspective dashboard layout that serves three distinct users (doctors, policymakers, and patients) simultaneously from a unified, locally hosted relational schema without relying on expensive cloud platforms.
4.	Social Impact / Customer Satisfaction	Saves doctors hours of manual data sorting, helps analysts back up healthcare policies with visual evidence, and reduces patient anxiety by giving them clear, color-coded tracking charts against health guidelines.
5.	Business Model (Revenue Model)	Open-source framework deployment utilizing free local tiers (MySQL Community and Tableau Desktop Public/Reader), eliminating ongoing software licensing fees for community hospitals and local departments.
6.	Scalability of the Solution	Built on an optimized MySQL InnoDB engine and Tableau Hyper extraction pipeline designed to automatically handle and smoothly refresh visual charts when new patient record rows are appended.